

Pawan Kumar Thapaliya

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Summary

A computational physicist and neurobiologist specializing in cellular neurobiology, computational neuroscience, and biophysics research. Expertise lies in mechanistic modeling of astrocyte-neuron interactions, ionic signaling, and energy metabolism under stroke and epilepsy. Highly skilled in differential equation and Markov-based simulations, integrating computational models with experimental data, and applying deep learning (CNNs, LSTMs, VAEs) to EEG and quantitative MRI data. Research interest includes the application of quantitative MRI and computational modeling approaches to glioblastoma. Passionate about translating complex neurobiological processes into predictive computational frameworks and innovative analytical pipelines.

Education

University of South Florida, Tampa, USA*Ph.D. in Applied Physics: Computational Biophysics and Neurobiology* **Aug 2019 – Aug 2026****University of Texas Rio Grande Valley**, Brownsville, Texas*M.S. in Physics: Computational Biophysics of DNA Stability* **Aug 2016 – Jul 2019****Central Department of Physics, Tribhuvan University**, Kathmandu, Nepal*MSc. in Physics: Solid State Physics and Biophysics* **Aug 2009 – Jul 2011****Trichendra College, Tribhuvan University**, Kathmandu, Nepal*BSc. in Physics: Physics, Statistics, and Mathematics* **Aug 2006 – Jul 2009**

Publications

1. **Thapaliya, P.**, Pape, N., Rose, C. R., and Ullah, G. (2023). *Modeling the heterogeneity of sodium and calcium homeostasis between cortical and hippocampal astrocytes and its impact on bioenergetics*. *Frontiers in Cellular Neuroscience*, 17, 1035553. Link — Code
2. Everaerts, K., **Thapaliya, P.**, Pape, N., et al. (2023). *Inward operation of sodium-bicarbonate cotransporter 1 promotes astrocytic Na^+ loading and loss of ATP in mouse neocortex during brief chemical ischemia*. *Cells*, 12(23), 2675. Link — Code
3. Hamzei Taj, S., **Thapaliya, P.**, Rakers, C., Gerkau, N. J., Rose, C. R., Ullah, G., Petzold, G. C. (2026) *Neuronal and astrocytic sodium-calcium exchanger differentially regulates calcium and sodium overload during ischemic stroke*. Submitted to *JCI Insight*. Link

Conference Presentations

- Symposium Talk: *Investigating the Reversal of GLT1 Glutamate Transporter during Metabolic Stress*, *Neuroscience 2024*, Chicago, IL.

- Poster: *Investigating the Reversal of GLT1 Glutamate Transporter during Metabolic Stress*, Neuroscience 2024, Chicago, IL.
- Poster: *Modelling the Impact of Astrocytic NBCe1 on Ischemia-Induced Astrocytic Na⁺ Loading and ATP Depletion*, Neuroscience Nov 2023, Washington, D.C.
- Poster: *Modeling heterogeneity of sodium and calcium homeostasis between cortical and hippocampal astrocytes*, USF Research Health Day Mar 2023, Tampa, FL.

Awards

- Trainee Professional Development Award, Society for Neuroscience, **Oct 2024**
- Signature Doctoral Research Fellowship, University of South Florida, **Aug 2024 – Jul 2026**
- Tharp and Duckwall Summer Research Fellowship, University of South Florida, **May 2024 – Aug 2024**

Experience

PhD Candidate / Researcher

University of South Florida, Tampa, FL

Aug 2019 – Aug 2026

Advisor: Ghanim Ullah, PhD

Key Achievements:

- Revealed region-specific astrocytic sodium and calcium dynamics by developing computational and biophysical models (FORTRAN/MATLAB/Python, differential equations, Markov models), showing cortical astrocytes have higher sodium transients and ATP consumption than hippocampal astrocytes.
- Demonstrated that functional differences in NMDA/AMPA receptors drive energy demand variations between cortical and hippocampal neurons through receptor modeling and analysis of fluorescence Imaging data.
- Characterized NBCe1's dual role in buffering ischemia-induced acidosis while accelerating sodium influx and ATP depletion using predictive simulations and comparative analysis.
- Identified contrasting effects of NCX1 in neurons versus astrocytes on sodium/calcium overload, glutamate release, and spreading depolarization thresholds via computational modeling validated with experimental data.
- Applied CNNs and LSTMs (Python, PyTorch, TensorFlow) to EEG and imaging datasets to detect ionic and metabolic disturbances, supporting model validation and biomarker identification.
- Published two first-author papers and presented at international conferences; maintained collaborations with Heinrich-Heine University and University of Bonn.

Quantitative MRI & Deep Learning Intern, Moffitt Cancer Center, Tampa, FL Jul 2025 – Dec 2025

Advisor: Dr. Tess Armstrong

- Investigated quantitative MRI research using diffusion phantoms and advanced MRI acquisition protocols for imaging optimization and biomarker development.
- Developed deep learning pipelines for image analysis, reconstruction, and diffusion metric evaluation.
- Utilized MRI processing and neuroimaging libraries including MONAI, NiBabel, pydicom, SimpleITK, and twixtools for multi-modal imaging workflows.
- Integrated advanced MRI techniques with ML workflows for reproducible analysis. Code

Research Fellow, University of South Florida

May 2024 – July 2026

Research Associate, University of South Florida

Summer 2023, Jan 2024 – May 2024

Deep Learning Training: PCA & VAE, Neuromatch Academy

Jul 2025

- Investigated the **intrinsic dimensionality** of MNIST and CIFAR-10 using **PCA and Variational Autoencoders (VAEs)**, testing latent dimensions from 10 to 500.
- Found that **VAEs produced more efficient representations than PCA**, achieving higher classification accuracy at lower latent dimensions.

Code

Computational Neuroscience Research, Neuromatch Academy

Jul 2024

Analyzed neural activity to uncover mechanisms of novelty detection and response variability in mice. Code

MS Researcher, University of Texas Rio Grande Valley, Brownsville, TX Aug 2016 – Jul 2019

Advisor: Andreas Hanke, PhD

- Developed **Monte Carlo simulations** to study short DNA duplexes, showing that **short DNA denatured mechanically while long DNA denatured thermally**.
- Generated **melting curves and force-extension profiles** and created a computational toolkit for DNA nanomechanics analysis.

Code

Research Assistant, University of Texas Rio Grande Valley, Brownsville, TX **Aug 2016 - May 2017, Summer (2018 and 2019)**

Investigated single-molecule interactions using DNA knot theory and Monte Carlo simulations to analyze DNA topology and structural dynamics. Built biophysical model to study mechanical and thermal stretching behaviors of helicoidal DNA.

Adjunct Instructor, University of South Florida, Tampa, FL

May 2020 - Jul 2023

- Taught lecture portions of General Physics II. Designed courses, graded, advised, and assisted approximately 150 undergraduate students.

Graduate Teaching assistant, University of South Florida, Tampa, FL **Aug 2019 - Dec 2023**

- Taught lecture and lab portions of General Physics I and Physics II. Designed courses, graded, advised, and assisted undergraduate students.

Graduate Teaching assistant, The University of Texas Rio Grande Valley, Brownsville, TX **Aug 2017 - July 2019**

- Taught lecture and lab portions of General Physics and Astro Physics. Designed courses, graded, advised, and assisted undergraduate students.

Technical Skills

- **Modeling and Mathematics:** Markov models, neural dynamics, receptor kinetics, ion channel modeling, predictive modeling, Runge-Kutta/Euler simulations, Monte Carlo, fluorescence image analysis.
- **Programming/Scripting Languages:** Python, R, Bash, MATLAB, SQL, FORTRAN.
- **Machine Learning/Frameworks:** Keras, scikit-learn, TensorFlow, PyTorch, SciPy, statsmodels.
- **Computing:** CUDA, Google Cloud Platform, Linux/Unix.
- **Code Development:** Git, GitHub, Anaconda.
- **Libraries:** NumPy, SciPy, MNE-Python, Biopython, OpenCV, NetworkX, Pandas, Matplotlib, Seaborn, Plotly Express.
- **MRI/Medical Imaging Libraries:** MONAI, NiBabel, pydicom, SimpleITK, twixtools, TorchIO, OpenCV, scikit-image.
- **Signal/Image Processing:** EEG analysis, image segmentation, modeling large neural circuits.

Leadership Experience

- Leader, SfN Neuronline Community, **Jan 2025 – Dec 2025**
- Secretary, Physics Graduate Student Committee, **May 2024 – May 2025**

Professional Activities

- Grand Award Judge, Regeneron International Science and Engineering Fair, **May 2024, 2025, 2026**
- Poster Judge, USF Health Research Day, **Mar 2024, Feb 2025**
- Reviewer: Cellular Signaling, Journal of Theoretical Biology

Licenses and Certificates

- SQL for Data Science, Coursera, **Sep 2020**
- Neural Networks and Deep Learning, Coursera, **Jul 2020**
- Python for Signal and Image Processing Master Class, Udemy, **Oct 2024**

Memberships

- Organization for Computational Neuroscience (OCNS), **2020 – Present**
- Society for Neuroscience (SfN), **2023 – Present**
- American Physical Society, **2016 – 2019**

- Golden Key International Honor Society, lifetime member, **Mar 2024**